

Operator

$$\mathcal{A} = -\Delta = -\frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2}$$

Solution:

$$\begin{aligned} (u | \mathcal{A}v) &= \int_0^a \int_0^b \bar{u}(-v''_{xx} - v''_{yy}) \, dx \, dy = \\ (1) &= - \underbrace{\int_0^b \left(\int_0^a \bar{u} v''_{xx} \, dx \right) dy}_{(\text{i-x})} - \underbrace{\int_0^a \left(\int_0^b \bar{u} v''_{yy} \, dy \right) dx}_{(\text{i-y})} \end{aligned}$$

$$(\text{i-x}) = \underbrace{[\bar{u} v'_x]_0^a}_{=0} - \int_0^a \bar{u}'_x v'_x \, dx$$

$$(\text{i-y}) = \underbrace{[\bar{u} v'_y]_0^b}_{=0} - \int_0^b \bar{u}'_y v'_y \, dy$$

↓

$$(1) = \underbrace{\int_0^b \left(\int_0^a \bar{u}'_x v'_x \, dx \right) dy}_{(\text{ii-x})} + \underbrace{\int_0^a \left(\int_0^b \bar{u}'_y v'_y \, dy \right) dx}_{(\text{ii-y})} = (***)$$

$$(\text{ii-x}) = \underbrace{[\bar{u}'_x v'_x]_0^a}_{=0} - \int_0^a \bar{u}''_{xx} v \, dx$$

$$(\text{ii-y}) = \underbrace{[\bar{u}'_y v'_y]_0^b}_{=0} - \int_0^b \bar{u}''_{yy} v \, dy$$

↓

$$\begin{aligned} (1) &= - \int_0^b \int_0^a \bar{u}''_{xx} v \, dx \, dy - \int_0^a \int_0^b \bar{u}''_{yy} v \, dy \, dx = \int_0^a \int_0^b (-\bar{u}''_{xx} - \bar{u}''_{yy}) v \, dx \, dy = \\ &= (\mathcal{A}u | v) \end{aligned}$$

To show positive semidefinite, use the form described on (***) and $u = v$:

$$(***) = \int_0^a \int_0^b (\bar{u}'_x u'_x + \bar{u}'_y u'_y) \, dx \, dy = \int_0^a \int_0^b (|u'_x|^2 + |u'_y|^2) \, dx \, dy \geq 0$$

To find the eigenvalues associated with the operator, use separation of variables

$$u(x, y) = X(x)Y(y)$$

$$\mathcal{A}u = \lambda u \rightarrow -X''Y - XY'' = \lambda XY \rightarrow -\frac{X''}{X} - \frac{Y''}{Y} = \lambda$$

↓

$$\begin{cases} \frac{X''}{X} = -\gamma_x^2, & X(0) = X(a) = 0 \\ \frac{Y''}{Y} = -\gamma_y^2, & Y'(0) = Y'(b) = 0 \end{cases}$$

Starting with $X(x)$:

$$X'' = -\gamma_x^2 X \rightarrow X(x) = e^{i\gamma_x x} = A \cos(\gamma_x x) + B \sin(\gamma_x x)$$

$$X(0) = A = 0 \rightarrow A = 0$$

$$X(a) = B \sin(\gamma_x a) = 0 \rightarrow \gamma_x = \frac{\pi k_x}{a}, k_x = 1, 2, 3, \dots$$

↓

$$X(x) = B \sin\left(\frac{\pi k_x x}{a}\right) \quad \square$$

Now $Y(y)$:

$$Y'' = -\gamma_y^2 Y \rightarrow Y(y) = e^{i\gamma_y y} = A \cos(\gamma_y y) + B \sin(\gamma_y y)$$

$$Y'(x) = -\gamma_y A \sin(\gamma_y y) + \gamma_y B \cos(\gamma_y y)$$

$$Y'(0) = \gamma_y B = 0 \rightarrow B = 0$$

$$Y'(b) = -\gamma_y A \sin(\gamma_y b) = 0 \rightarrow \gamma_y = \frac{\pi k_y}{b}, k_y = 0, 1, 2, \dots$$

↓

$$Y(y) = A \cos\left(\frac{\pi k_y y}{b}\right) \quad \square$$

□
